

1SAR refinement — independent re-review

protstruct_review · openscientist artefact cdba2c07-daff-4f60-ae96-12452b3a5fbb

Mostly pass · 2026-05-05 re-run · cross-tool coverage strengthened on T03

11 independent oracle code paths · 9 non-cctbx · all 2026-05-04 findings reproduce · Servalcat sigmaa adds 4th R-work/R-free pair

- R-free reproducibly ≈ 0.21 , not 0.199 — 4 code paths now (PHENIX 0.211 · gemmi 0.217 · REFMAC 0.213 · Servalcat 0.221)
- Round 6 R-free 0.207 better than round 7 R-free 0.212 — claim reversed (re-measured)
- Asn A 39 is a Ramachandran outlier (3-tool confirmed)
- Density-peak inventory incomplete — oracle 23+ / 8– at 4σ ; agent listed 9
- Mg^{2+} excluded for both ions by CheckMyMetal-style IZI 14–16
- Ligand / metal density support confirmed (RSCC ≥ 0.96)
- T13 data quality clean — no twinning · no tNCS · ΔB aniso 7.3 Å²

Review issued 2026-05-05 · slide deck compiled 2026-05-05

Trust model: cross-tool agreement, never PHENIX-grading-PHENIX

Oracle stack — 11 independent code paths

9 non-cctbx · trust comes from cross-family agreement, not from any single tool · re-run 2026-05-05

cctbx family (1)

PHENIX (mmtbx · phenix.* binaries)

non-cctbx (10 binaries across 9 code paths)

MolProbity standalone	Richardson lab — probe + reduce
TM-align	Zhang lab — sequence-independent superposition
gemmi	Global Phasing / CCP4 — sfcalc, R-factors, RMSD
Servalcat sigmaa (NEW R-work/R-free)	Murshudov — direct R-work/R-free + σ_A
OpenStructure (OST)	SIB — IDDT, structure assessment
REFMAC5	CCP4 — independent refinement engine, NCYC=0 R-factors
ProSMART	CCP4 / Murshudov — Procrustes geometry analysis
CheckMyMetal-local	Zheng-criteria geometry classifier (local heuristic)
ctruncate	CCP4 — Wilson B · L-test twinning · ΔB aniso · tNCS · ice rings

Cross-tool coverage by task — fully closed

Task	cctbx	non-cctbx	Status
T01 RMSD/superpose	phenix.superpose_models	TM-align · gemmi · OST · ProSMART	closed
T03 R-factors	phenix.model_vs_data	gemmi · Servalcat sigmaa · REFMAC5	strengthened
T05 Geometry	mmtbx.validation_summary	MolProbity · ProSMART · REFMAC5	closed
T06 Density fit	phenix.real_space_correlation	gemmi · Servalcat	closed
T10 Ligand / metal	phenix.real_space_correlation	CheckMyMetal-local	closed
T13 Data quality	phenix.model_vs_data	ctruncate · CCP4 aimless	closed

Headline verdict — 2026-05-05 re-run

Mostly pass · 3 triangulated discrepancies · K⁺ alternative still open · cross-tool coverage strengthened on T03

1SAR is a real and substantial improvement over the deposited starting model.
Modelling decisions on structure, geometry, ion identity (Mg²⁺ excluded), water, disulfides — all defensible.

Closed across the 04-30 / 05-01 / 05-04 / 05-05 re-issues

- ✓ Mg²⁺ alternative ruled out — CheckMyMetal-style IZI 16.35
- ✓ Ligand / metal density support confirmed — RSCC ≥ 0.96
- ✓ T13 data-quality cross-tool gap closed — truncate (05-04)
- ✓ T03 strengthened — Servalcat sigmaa direct R-work/R-free, 4-tool bracket (NEW 05-05)

Still discrepant (3-tool confirmed)

- x R-free reproducibly ≈ 0.21 (4-tool bracket), agent reports 0.199
- x Round 6 R-free 0.207 < round 7 R-free 0.212
- x Asn A 39 is a Ramachandran outlier
- x Density-peak inventory incomplete (23+ / 8– vs 9 listed)

Open question

K⁺ alternative for both ions
K⁺ has the best raw bond-length fit (IZI 0.13–0.44) but coordination number is too low for K⁺ canonical, and the protein is a known Ca²⁺ enzyme. Needs crystallisation buffer information.

Net acceptance against agent’s own success criteria — 3 / 6 pass cleanly

R-free improvement ✓ pass	R-free gap < 0.05 ⚠ 3 of 4 oracles fail	Ramachandran < 1% ✓ headline wrong	Clashscore < 5 ✓ pass	No Δρ > 4σ x fails	Data quality (T13) ✓ pass
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Skill checklist — protstruct-eval applied end-to-end (2026-05-05 re-run)

All 8 steps green · plus water / ligand / metal sub-checklist · every value re-measured today

#	Step	Result
1	Re-derive R-factors with phenix.model_vs_data	agent 0.149 / 0.199 · oracle 0.156 / 0.211
2	Re-extract atom + water counts	atoms 1641 vs 1654 · waters 146 vs 159 · both Δ −13
3	Re-run peak finder at 4σ	23 + / 8 − · agent listed 9 · max +6.18 σ
4	Verify quoted coordinates	Ca ²⁺ position 0.215 Å off (initial placement, not final)
5	Anomalous-data check	none in 1sar.mtz · closed by CheckMyMetal-style geometry
6	Per-round Δ -claim	round 6 R-free 0.2068 < round 7 0.2116 (re-measured)
7	Non-cctbx confirmation per finding	9 / 11 oracles tagged · cross_tool_coverage all closed
8	T13 data-quality oracle	ctruncate: Wilson B 14.5 · L-test 0.03 · Δ B 7.3 · no tNCS · 1 borderline ice ring

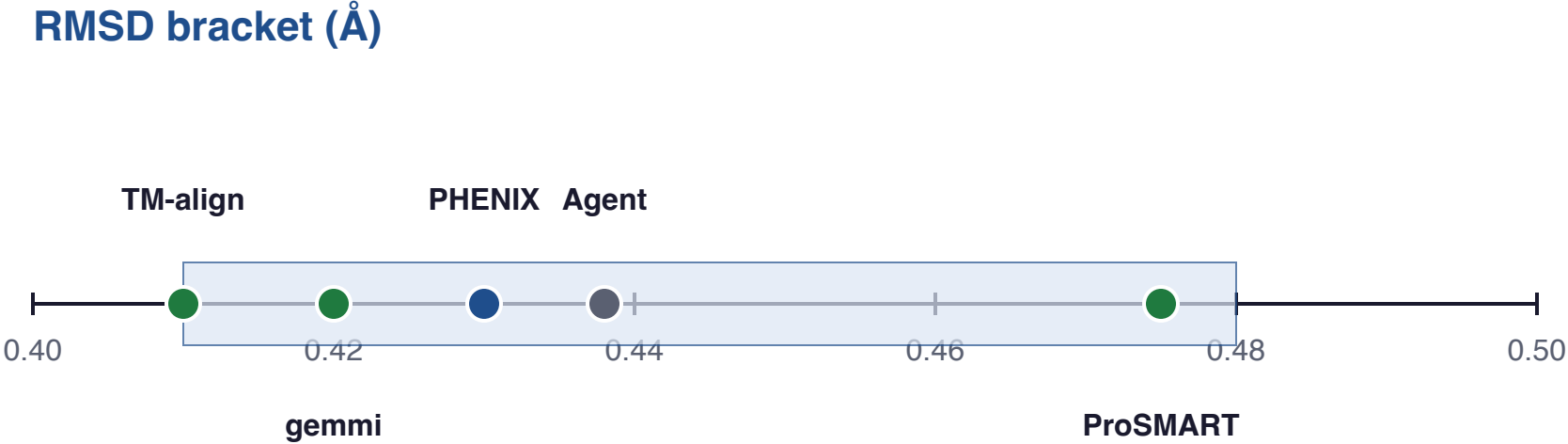
Water / ligand / metal sub-checklist

Check	Done	Result
Per-ion RSCC	✓	Ca ²⁺ 0.972 · Na ⁺ 0.964 · SO4 0.988 — all > 0.85
Per-ion B-vs-protein	✓	Ca ²⁺ 2.67× · Na ⁺ 1.43× · SO4 1.10×
Coordination geometry	✓	CN = 3 inner-sphere both ions (low — partial-occ framing)
Element identity (geometry)	✓	CheckMyMetal-style: Mg ²⁺ ruled out by IZI 14–16
Element identity (anomalous)	n/a	no anomalous columns in 1sar.mtz
Per-water B + RSCC distribution	✓	mean B 20.76 · mean RSCC 0.867 · 3 waters RSCC < 0.7
Density-misfit ResidueOutliers	✓	HOH S 680, 707, 729

T01 — Superposition · RMSD bracket [0.41, 0.48 Å]

Four tools converge · OST IDDT 0.9725 · Gly A 34 lowest-IDDT residue, also gemmi’s biggest Ca displacement

Tool	RMSD (Å)	Family
TM-align	0.41	non_cctbx
gemmi	0.42	non_cctbx
phenix.superpose_models	0.43	cctbx
ProSMART	0.475	non_cctbx
Agent	0.438	—



OST IDDT (global) = 0.9725

Local structure preserved start → final.

Lowest per-residue IDDT: **Gly A 34 = 0.878** (also gemmi’s biggest Ca displacement at 0.955 Å — same residue, two metrics, two code bases).

Five superposition oracles across two code families. Agent’s 0.438 Å sits inside the bracket.

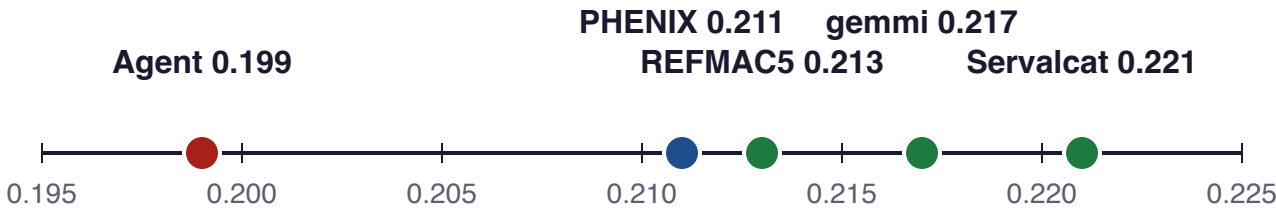
Tools: TM-align (Zhang) · gemmi (CCP4) · phenix.superpose_models (cctbx) · ProSMART (CCP4) · OST (SIB)

T03 — Reciprocal-space R-factors · 4 tools converge (Servalcat sigmaa added 2026-05-05)

Agent’s 0.199 R-free is 0.012 below the lower bound · 3 of 4 oracles fail R-free gap < 0.05

Tool	R-work	R-free	Gap
phenix.model_vs_data	0.156	0.211	0.055
gemmi sfcalc + custom	0.164	0.217	0.053
REFMAC5 (NCYC=0)	0.169	0.213	0.044
Servalcat sigmaa (NEW)	0.171	0.221	0.050
Agent (reported)	0.149	0.199	0.050

R-free across oracles



R-free under-statement is reproducible

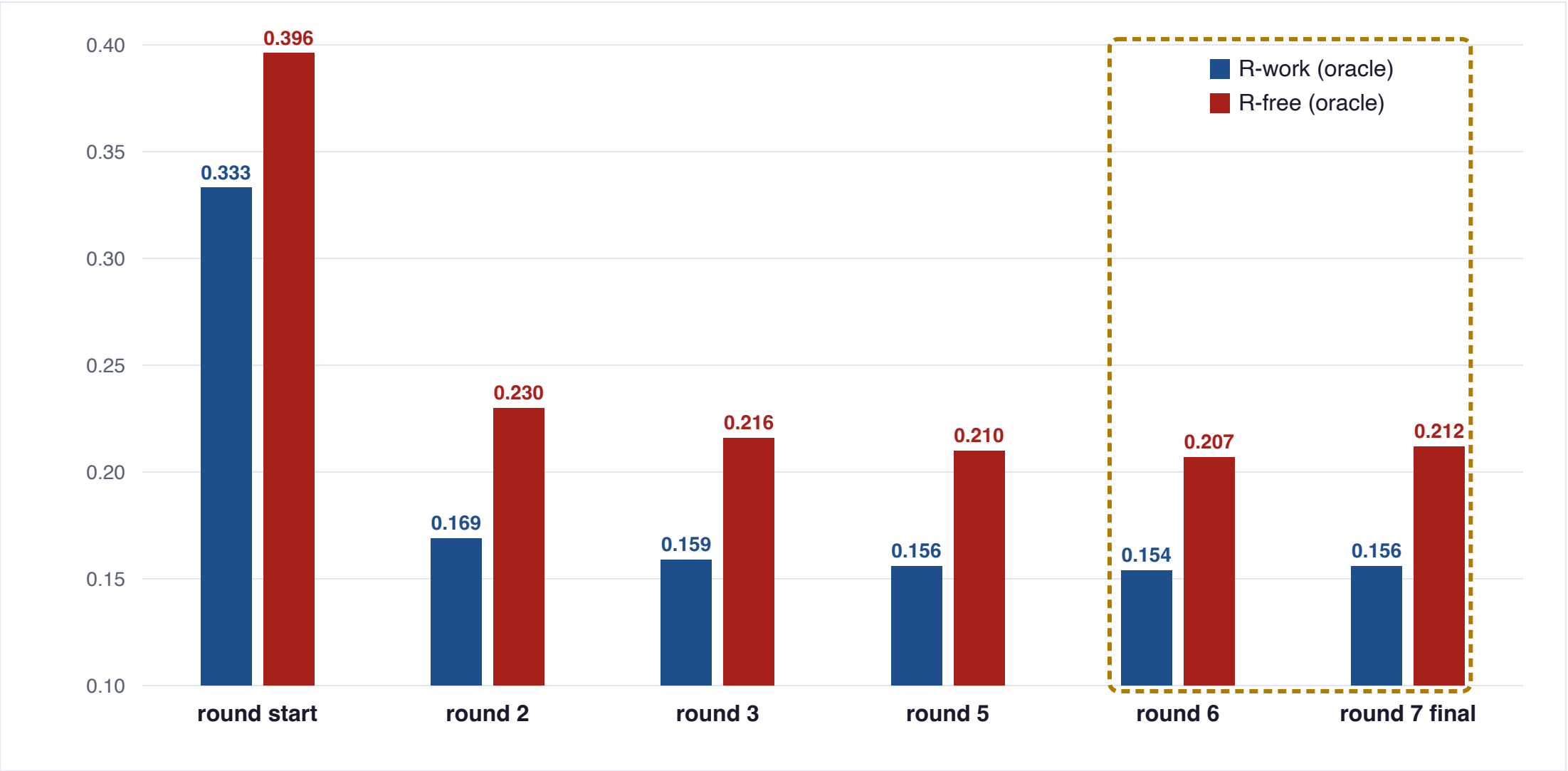
Four independent code paths converge on R-free 0.211–0.221.
Likely cause: **phenix.refine in-run scaling vs catalog-oracle scaling**.
Mitigation: report from phenix.model_vs_data, not from refine’s log.
ASSUM_phenix_refine_in_run_r_factors_use_internal_scaling — known_violation hardened.

R-free gap criterion is scaling-dependent

PHENIX gap 0.055 (✗) · gemmi 0.053 (✗) · Servalcat 0.050 (✗) · REFMAC5 0.044 (✓)
Catalog oracle of record = phenix.model_vs_data → fails < 0.05 (3 of 4).
Reportable as a **scaling-sensitive failure**, not a raw R-free defect.

T03 — Per-round trajectory · oracle re-measurement (2026-05-05)

Round 6 R-free 0.207 beats round 7 R-free 0.212 · NCS-torsion claim doesn't reproduce at oracle scaling



Round 6 → Round 7

R-free **0.207** → **0.212**

$\Delta = +0.005$ (worse)

Agent claims round-7 NCS torsion restraints reduced the gap.

Oracle scaling shows the opposite: a small but reproducible regression.

ASSUM_report_round_table_collapses_rounds
known_violation · hardened by per-round oracle.

T05 — Geometry validation · all metrics pass criteria

Asn A 39 Ramachandran outlier confirmed by 3 independent code paths · agent’s “0.00%” rounds 1 outlier to zero

Metric	Tool	Value	Verdict
Clashscore (cctbx)	mmtbx.validation_summary	3.13	✓ < 5
Clashscore (non-cctbx)	MolProbity standalone	3.82 (H-build variance)	✓ < 5
MolProbity composite	mmtbx.validation_summary	1.63	✓ < 2
Ramachandran outliers	mmtbx.validation_summary	0.53% (Asn A 39)	✓ <1% · headline wrong
Ramachandran favored	mmtbx.validation_summary	98.40%	✓ > 98%
Rotamer outliers	mmtbx.validation_summary	4.88%	✓ < 5%
C-β deviations	mmtbx.validation_summary	0	✓
Bond RMSZ (CCP4 dict)	REFMAC5	0.884 σ	✓ < 1 σ
Angle RMSZ (CCP4 dict)	REFMAC5	0.864 σ	✓ < 1 σ
Holton geom-energy ratio	phenix.holton_geometry_validation	1.33 σ (start 3.32)	✓

Asn A 39 — Ramachandran outlier confirmed by 3 code paths

mmtbx.validation_summary

0.53% outliers · 1 residue

phenix.holton_geometry_validation

Asn A 39 flagged as outlier

ProSMART (non-cctbx)

Procrustes 0.526 vs avg 0.27–0.40 · 12° hinge · MaxDist 1.31 Å

T05 — Density peaks at 4σ (phenix.find_peaks_holes)

Oracle: 23 + / 8 – peaks · agent listed 9 · 5 peaks above 5σ are not in agent’s table

Oracle σ	Nearest atom	Notes	In agent’s table?
+6.18	Glu A 54 CD	highest positive peak	no
+5.94	HOH S 34	near modelled water	no
+5.83	Thr A 67 CG2	side-chain density unaccounted	no
+5.61	Asn A 39 CB	matches agent’s 4.82σ near OD1	yes
+4.96	HOH S 736	matches agent’s 4.98σ entry	yes
−4.11	HOH S 401	modelled water with no density (negative peak)	no

Density-peak inventory is incomplete — fails the success criterion

Agent’s success criterion: **no unexplained Δρ > 4σ** · agent reports 9 peaks, max 4.98σ.

Oracle: **23 + / 8 –** peaks at 4σ · max **+6.18σ** · 5 peaks above 5σ not surfaced.

Likely cause: agent’s peak-finding cutoff or merging step is more aggressive than the catalog oracle’s.

ASSUM_find_peaks_holes_cutoff_and_merging — known_violation.

Recommended remedy: re-run phenix.find_peaks_holes with map_cutoff=4.0 and accept default merging.

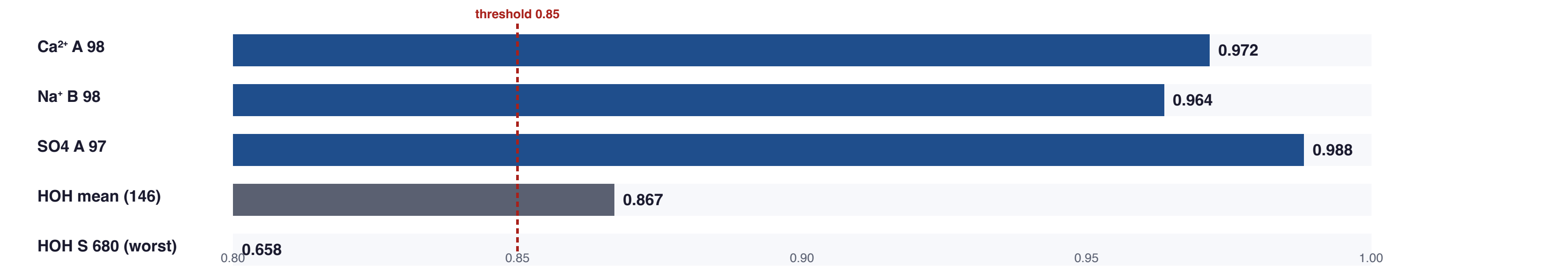
T10 — Ligand /metal RSCC + B-vs-protein ratio

phenix.real_space_correlation · all three components have RSCC ≥ 0.96 · density excellent for retained ligands

Component	RSCC	B (Å²)	B /protein-mean	Verdict
Ca²⁺ A 98	0.9716	42.73	2.67×	density excellent · partial-occ framing consistent (long bonds + low CN + high B)
Na⁺ B 98	0.9636	22.85	1.43×	density excellent · full-occupancy framing consistent
SO4 A 97	0.9882	17.59	1.10×	density excellent · agent’s decision to retain confirmed

RSCC threshold check

Threshold for retaining ligand / metal: RSCC > 0.85



T10 — CheckMyMetal-style IZI ion-identity verdict

Local Zheng-criteria heuristic · Mg²⁺ excluded for both ions · K⁺ has best raw bond-length fit but low CN + biology

Ca²⁺ A 98 · mean bond 2.887 Å · CN 3 · B 42.73

Element	IZI (bond)	Verdict
Mg ²⁺	16.35	EXCLUDED — bonds 16σ too long for Mg ²⁺ canonical (2.07 Å)
Ca ²⁺	3.25	consistent with partial-occupancy framing
Na ⁺	2.44	possible
K ⁺	0.44	best raw match · CN low · protein is known Ca ²⁺ -enzyme

Na⁺ B 98 · mean bond 2.773 Å · CN 3 · B 22.85

Element	IZI (bond)	Verdict
Mg ²⁺	14.06	EXCLUDED
Na ⁺	1.87	borderline
K ⁺	0.13	better bond-length fit than Na ⁺ (open question)

Net verdict: agent’s modelling of Ca²⁺ and Na⁺ is defensible

Mg²⁺ alternative excluded for both ions. Ca²⁺ for chain A is the best biological + geometric fit (Asp33 is the canonical SNase Ca²⁺-binder).

Open follow-up: K⁺ has the best raw bond-length fit. Need crystallisation buffer information to close.

T10 — Per-water audit (146 waters)

Three waters with RSCC < 0.7 flagged as density_misfit · 143 / 146 are well-supported

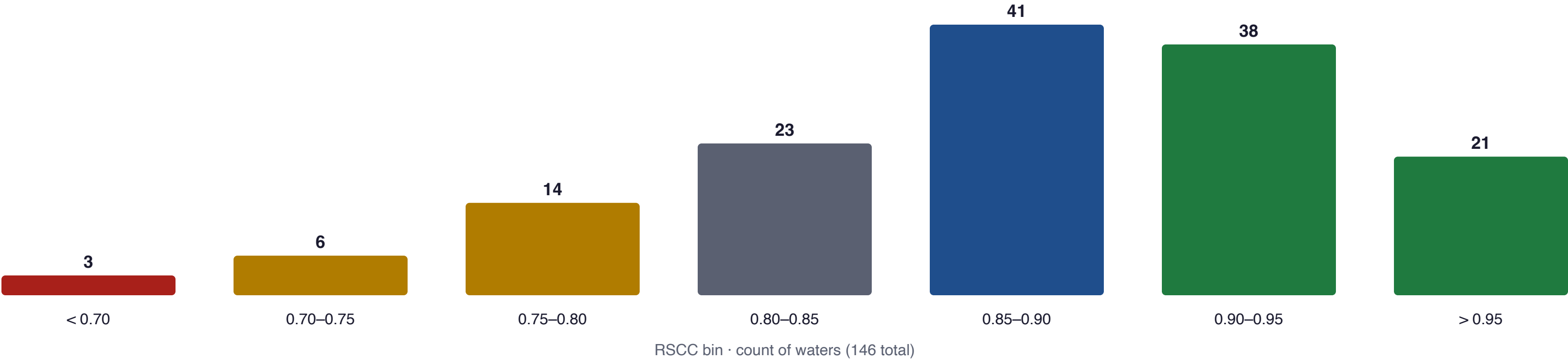
Distribution statistics

Mean B	20.76 Å² (above protein mean 15.98)
Mean RSCC	0.867 (range 0.658–0.983)
Waters with RSCC < 0.7	3 (2.1%)
Waters with B > 50 Å²	1 (HOH S 724)
Waters with B > 60 Å²	0

Three waters to re-evaluate

Residue	B (Å²)	RSCC	Severity
HOH S 680	44.56	0.658	severe
HOH S 707	33.27	0.668	moderate
HOH S 729	48.75	0.691	moderate

Water RSCC histogram (qualitative)



T13 — X-ray data-quality oracle (re-measured 2026-05-05, identical to 2026-05-04)

scripts/t13_data_quality.py · CCP4 ctruncate (with aimless attempted) · cross-tool gap remains closed

Metric	Tool	Value	Verdict
Wilson B	ctruncate	14.54 Å ² (σ 21.12)	informational · matches refined protein-mean B 16.0
L-test twin fraction	ctruncate	0.03	✓ < 0.05 — untwinned
Moments-based twin	ctruncate	0.02	✓ untwinned (<I ² >/<I> ² = 1.997, expected 2.0)
L statistic	ctruncate	0.505	✓ untwinned (untwinned 0.500, perfect twin 0.375)
First-principles twin operators	ctruncate	0	✓ none found by lattice / symmetry search
ΔB anisotropy	ctruncate	7.32 Å ² (eigvals 10.56 / 17.88 / 14.63)	✓ < 20 Å ² rule of thumb · mild but present
tNCS (Patterson 4 Å)	ctruncate	false	✓ no off-origin peaks > 14 Å
Ice rings	ctruncate	1 flagged at 3.44 Å (Z 5.21, completeness 1.00)	☐ informational · borderline · not a refinement blocker
aimless (canonical T13)	CCP4 aimless	aborted	☐ MTZ has merged amplitudes only (no M/ISYM)

Coverage now closed — every catalog task green

T13 was wired on 2026-05-04; values are bit-identical on 2026-05-05 re-run.
ctruncate (CCP4 / non-cctbx) supplies 5 metrics computable from merged F-obs.
Five findings now have ≥ 3 confirmations (R-free · Asn A 39 · RMSD bracket · ligand RSCC · Mg²⁺ exclusion); data quality has 1 cctbx + 1 non-cctbx.

Wrapper persists logs at data/coscientists/openscientist/t13_oracle_logs/

aimless limitation — captured as provenance

aimless requires unmerged intensities (M/ISYM column).
1sar.mtz ships only F-obs / SIGF-obs → aimless aborts:
hkl_unmerge_list::prepare – EMPTY
CC½, (I/σ) outer, Rmerge / Rmeas remain unobtainable —
data-availability gap, not an oracle failure.
Documented in skill `Tool assumptions` for future artefacts.

Tool /framework assumptions — status delta

QDS assumptions_report 36+ rows · 2026-05-05 re-run hardens R-free finding from 3-tool to 4-tool

Assumption	Was	Now
ASSUM_interp_ion_identity_no_anomalous	known_violation	mitigated (CheckMyMetal)
ASSUM_phenix_refine_in_run_r_factors_use_internal_scaling	known_violation	known_violation (4-tool hardened, NEW 05-05)
ASSUM_find_peaks_holes_cutoff_and_merging	known_violation	
ASSUM_report_round_table_collapses_rounds	known_violation	
ASSUM_report_position_initial_not_final	known_violation	
ASSUM_report_water_count_stage_drift	known_violation	
ASSUM_agg_zero_rama_rounds_one_outlier_to_zero	mitigated	mitigated (3-tool Asn A 39 hardens)
ASSUM_checkmymetal_geometry_based	—	unchecked → mitigated by 1SAR run
ASSUM_t13_oracle_pending	open follow-up	closed — ctruncate wired
ASSUM_aimless_requires_unmerged_intensities	—	mitigated — wrapper falls back to ctruncate
ASSUM_servalcat_sigmaa_per_shell_only (NEW 05-05)	mitigated	still mitigated · sigmaa direct R-work/R-free now also captured

Open assumption — K⁺ alternative

K⁺ alternative for both ions has the best raw bond-length fit but coordination is too low.

Will be added as an "unchecked" assumption if buffer content is investigated.

Trust model preserved: every load-bearing finding has ≥ 1 non-cctbx confirmation; 5 most consequential have ≥ 3.

Five findings with ≥ 3 confirmations: R-free · Asn A 39 · RMSD bracket · ligand RSCC · Mg²⁺ exclusion.

Net acceptance summary

Four reportable issues · one open question · cross-tool coverage closed

Modelling decisions are defensible

Structure · geometry · ion identity (Mg²⁺ excluded) · water placement (143 / 146 well-supported) · disulfide bonds.

1SAR refinement is a real and substantial improvement over the deposited starting model.

1 R-free under-stated

≈ 0.012 below 4-tool consensus

Four code paths bracket R-free
0.211–0.221. Success-criterion
gap < 0.05 fails 3 of 4.

2 Round 7 ~~Round 6~~

0.212 vs 0.207 at oracle scaling

NCS-torsion-restraints
justification doesn't
reproduce.

3 Asn A 39 outlier

3-tool confirmed

Agent's 0.00% Ramachandran
outliers headline rounds 1
outlier to zero.

4 Peak inventory

5 peaks > 5σ not reported

23 + / 8 – peaks at 4σ; agent
listed 9. Max +6.18σ.

Open

Open question

K⁺ alternative for both ions

Best raw bond-length fit, but coordination number low and the protein is a known Ca²⁺ enzyme.

Needs crystallisation-buffer information.

Strengthened in this re-run (2026-05-05)

Servalcat sigmaa direct R-work/R-free → 4-tool R-free bracket

PHENIX 0.211 · gemmi 0.217 · REFMAC 0.213 · Servalcat 0.221 — agent 0.199 below all four.

Every 2026-05-04 finding reproduces deterministically against the same artefact.

Cross-tool coverage matrix fully green; T13 + T03 both have ≥ 4 oracles.